

Aegis Prototype Feature Inventory

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This document records the capabilities currently implemented in the Aegis workstation prototype. The end-to-end demonstration path is operational: a local user can upload and read material, ask a source-grounded question, run an inspection workflow, download editable Office files, verify a small utility in isolation and inspect local connection observations.

Current validation

32	3	5	3
Automated tests passed	Local model roles	Dashboard views	Editable Office formats

Implemented features

Area	Capability	State
Local identity	Local account creation, secure password hashing, expiring sessions, request protection, sign-out and authentication rate limiting.	Complete
Private workspaces	Per-user workspaces with ownership checks for materials, answers, workflows and deliverables.	Complete
Dashboard	Overview metrics, activity, desktop and mobile navigation, onboarding guide and compact display mode.	Complete
Material upload	PDF, TXT, CSV, PNG and JPEG by picker or drag and drop, with per-file and workspace limits.	Complete
Material explorer	Filename search, readiness filters, downloads, file-type labels and local-storage status.	Complete
Document reading	Text extraction, scanned-page OCR, page previews, extracted-text review and accuracy warnings.	Complete
Local model routing	General, coding and vision roles selected automatically from the task, with no cloud fallback.	Complete
Grounded answers	Local retrieval, page-linked sources, citation checks and an extractive fallback when model references fail.	Complete
Multimodal review	Local image interpretation for images and the first page of scanned PDFs.	Complete
Inspection workflow	Read, retrieve, draft, check references, retry once and create editable outputs.	Complete
Office deliverables	Approval note in DOCX, findings register in XLSX and briefing in PPTX.	Complete
Verified utilities	Fixed-purpose code generation, three independent cases, one repair attempt and downloadable source.	Complete

Area	Capability	State
Execution isolation	WebAssembly and WASI execution with time, fuel, memory and output limits and no host-directory or network grants.	Complete
Workflow history	Live status, progress, event history, verification checks, filters and grouped downloads.	Complete
Offline visibility	Loopback binding, Python outbound guard and application plus child-process connection observations.	Prototype evidence
Persistence and audit	SQLite persistence for project records and local audit events.	Complete

Current limitations

The following items define the boundary of the current demonstration and should remain visible during evaluation.

- DOCX, XLSX and PPTX are output formats and are not accepted as source uploads yet.
- Material and workspace deletion controls are not present.
- Retrieval is lexical and sized for a demonstration collection rather than a large enterprise corpus.
- The application database is not encrypted, and a Windows administrator can access workstation files.
- Password recovery, administrator provisioning, invitations and enterprise SSO are not included.
- Connection snapshots can miss short-lived traffic; packet-level capture is not yet complete.
- Generated text, OCR, vision, calculations and engineering interpretations require human review.
- The workstation configuration permits one active model generation request at a time.

Recommended demonstration sequence

- 1. Sign in and open a private workspace.** Show that another account cannot see its records.
- 2. Upload and read the fictional inspection scan.** Open the page image and extracted OCR text together.
- 3. Ask a source-grounded question.** Open the page-linked evidence from the saved answer.
- 4. Create an inspection review pack.** Follow workflow events and download the DOCX, XLSX and PPTX outputs.
- 5. Generate and verify a utility.** Show the three independent checks and downloadable source file.
- 6. Open Offline status.** Show the loopback binding and current connection observations with the stated evidence limit.

Prototype status: Ready for a controlled SIH demonstration using public or fictional material. Human review remains required for every generated conclusion and deliverable.